EK - MO 10. Frekvenční charakteristiky pasivních jednobranů

Pro získání poznatků o závislosti modulu a fáze impedance na frekvenci normujeme impedanci oproti rezistoru (abychom měli „jak moc je impedance větší/menší než rezistor nebo o kolik dB je impedance větší/menší než rezistor“)

A) RC

a) série

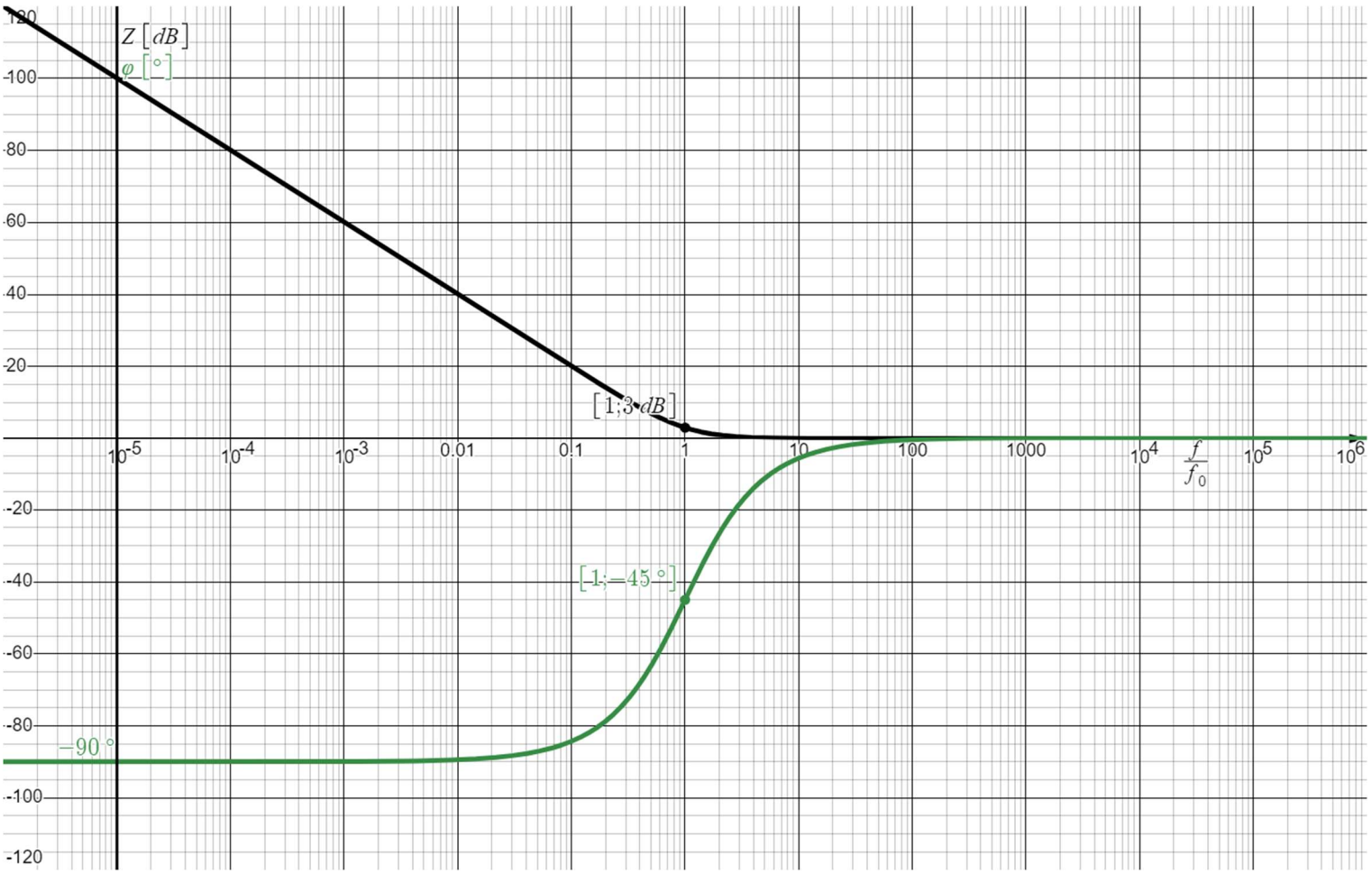
$$\hat{Z} = R + \frac{1}{j\omega C} = \frac{1 + j\omega RC}{j\omega C}$$

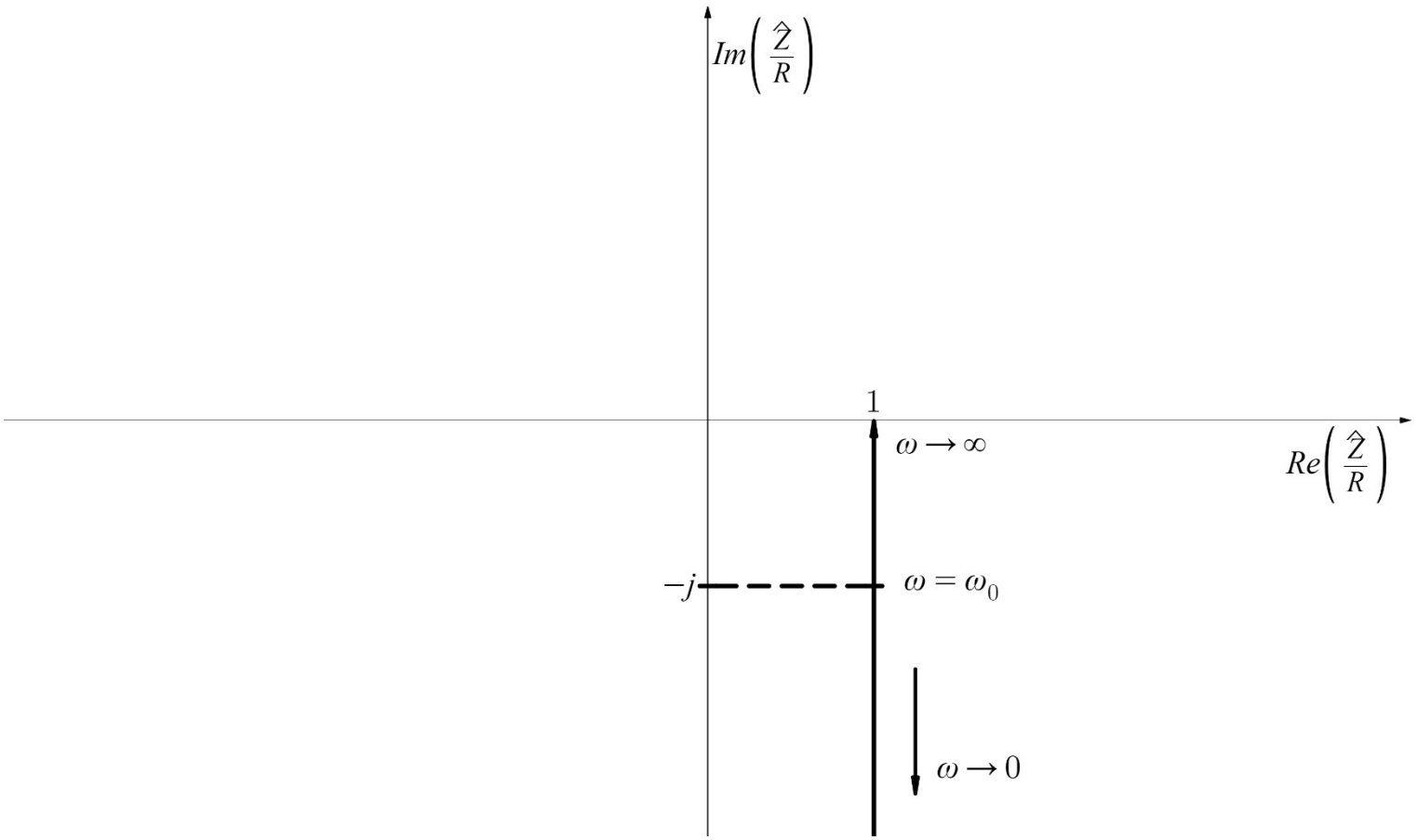
$$\tau_0 = RC \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{RC} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi RC}$$

$$\frac{\hat{Z}}{R} = \frac{1 + j\omega RC}{j\omega RC} = \frac{1 + j\omega\tau_0}{j\omega\tau_0}$$

$$\hat{Z}_{dB} = 20\log\left|\frac{\hat{Z}}{R}\right| = 20\log\left|\frac{1 + j\omega\tau_0}{j\omega\tau_0}\right|$$

$$\varphi = \arg\left(\frac{\hat{Z}}{R}\right) = \arg\left(\frac{1 + j\omega\tau_0}{j\omega\tau_0}\right) = \arg(1 + j\omega\tau_0) - \arg(j\omega\tau_0) = \operatorname{arctg}(\omega\tau_0) - \frac{\pi}{2}$$





b) paralel

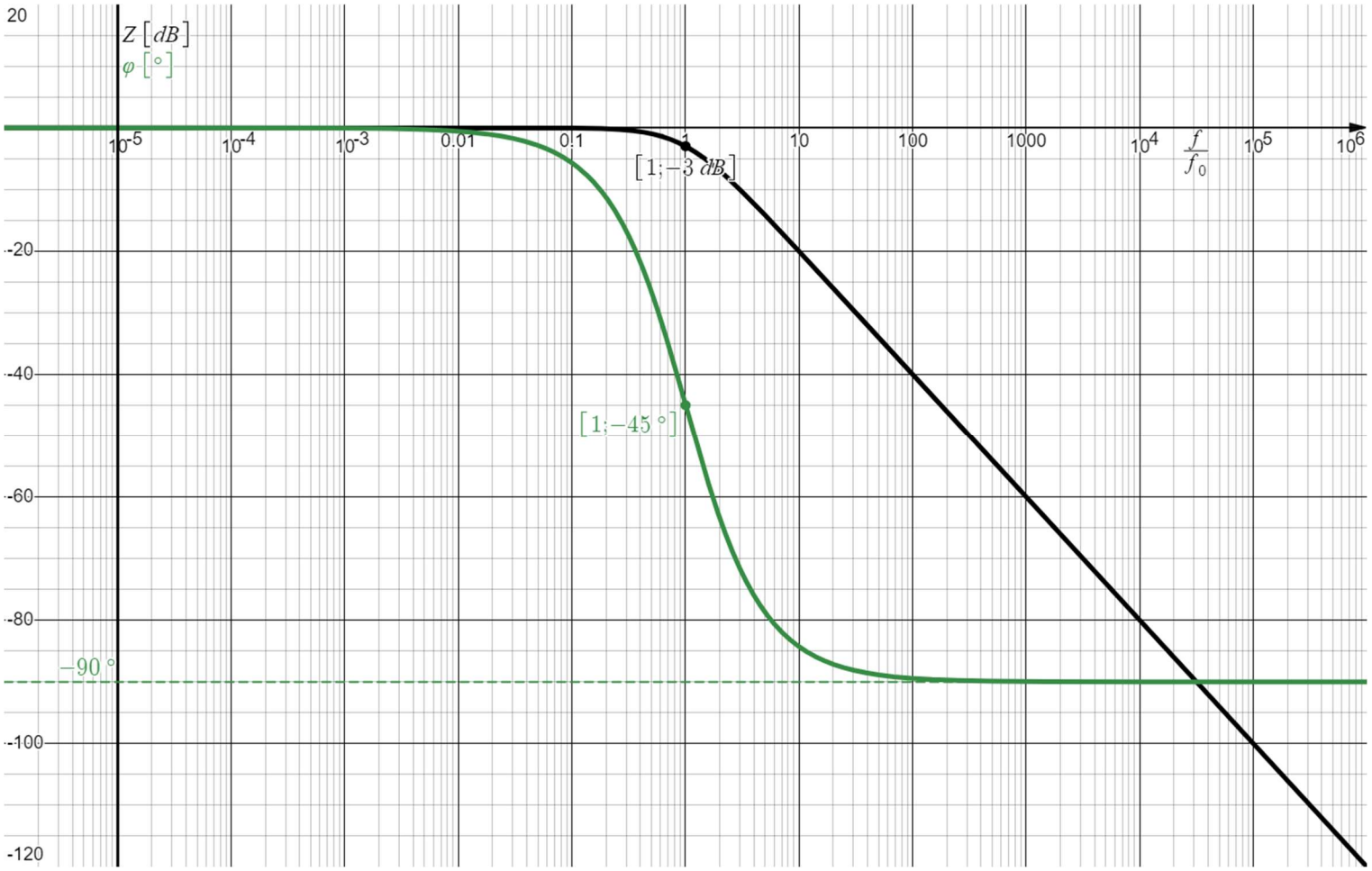
$$\hat{Z} = \frac{R \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \cdot \frac{j\omega C}{j\omega C} = \frac{R}{1 + j\omega RC}$$

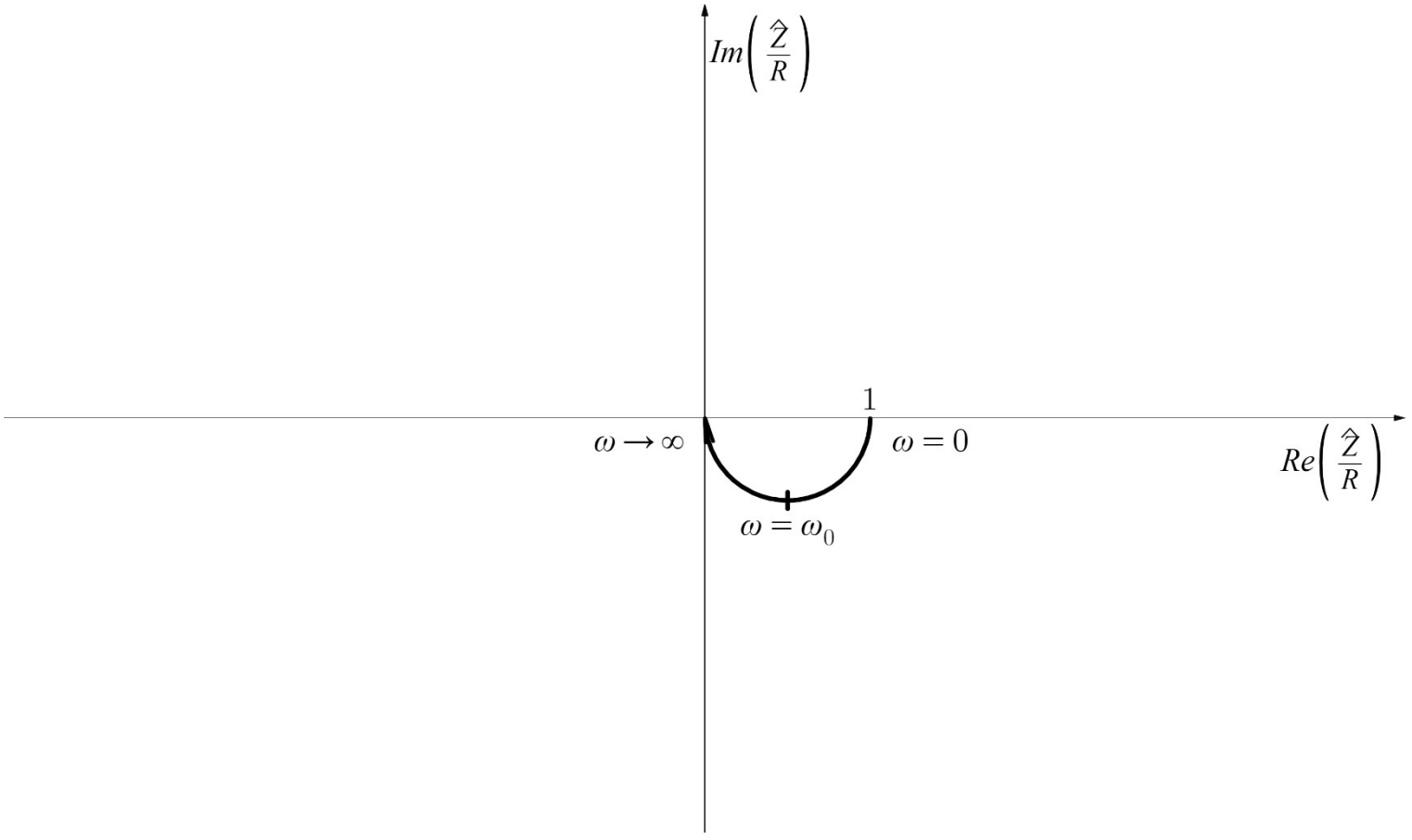
$$\tau_0 = RC \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{RC} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi RC}$$

$$\frac{\hat{Z}}{R} = \frac{R}{R(1 + j\omega RC)} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j\omega\tau_0}$$

$$\hat{Z}_{dB} = 20\log\left|\frac{\hat{Z}}{R}\right| = 20\log\left|\frac{1}{1 + j\omega\tau_0}\right|$$

$$\varphi = \arg\left(\frac{\hat{Z}}{R}\right) = \arg\left(\frac{1}{1 + j\omega\tau_0}\right) = \arg(1) - \arg(1 + j\omega\tau_0) = 0 - \operatorname{arctg}(\omega\tau_0)$$


20
Z [dB]
φ [°]
10-5
10-4
10-3
0.01
0.1
1
10
100
1000
10^4
10^5
10^6
f/f0
[1;−3 dB]
-20
-40
[1;−45 °]
-60
-80
−90 °
-100
-120


Im(Ẑ/R)
1
ω → ∞
ω = 0
Re(Ẑ/R)
ω = ω0

B) RL

a) série

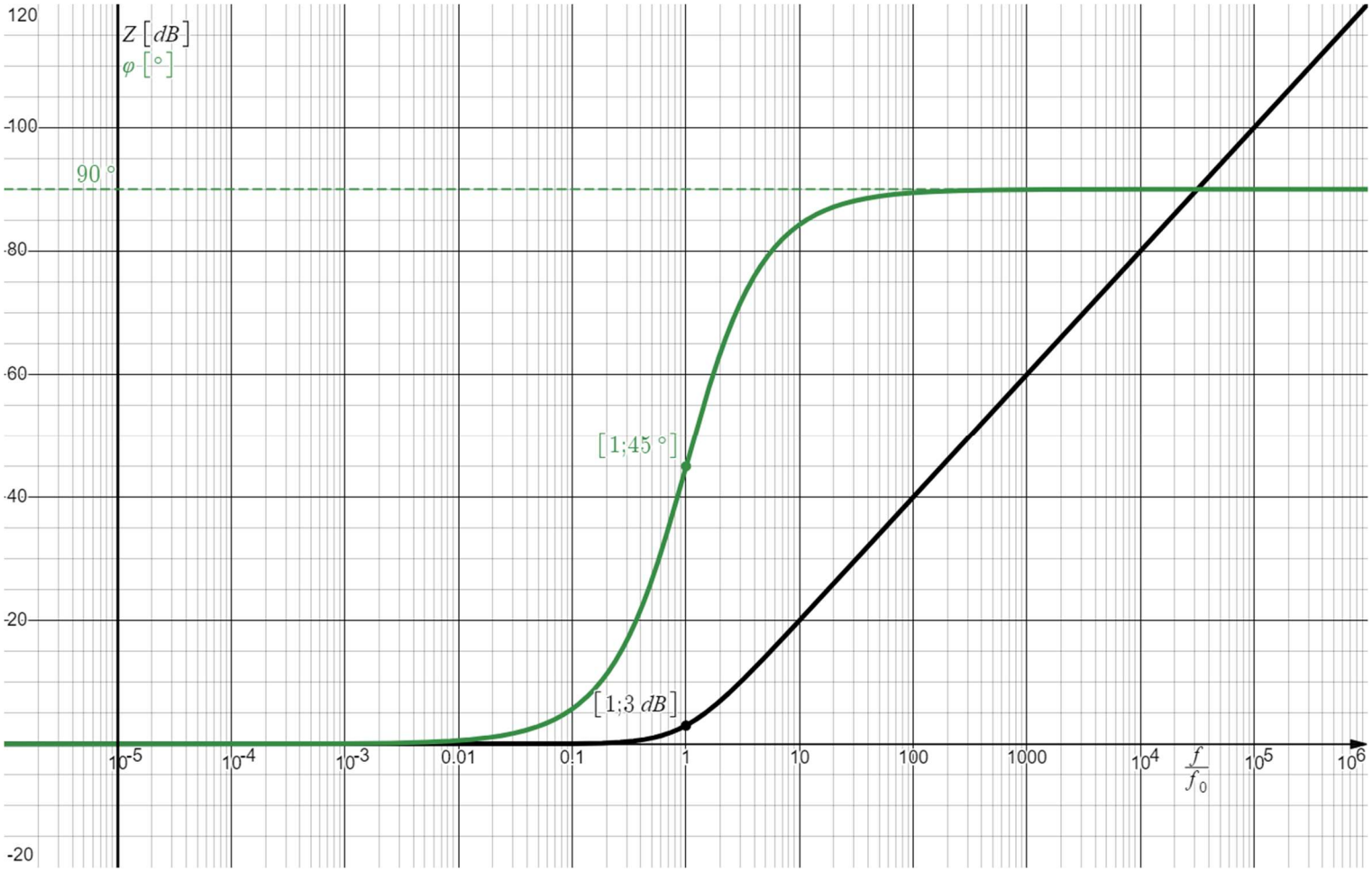
$$\hat{Z} = R + j\omega L$$

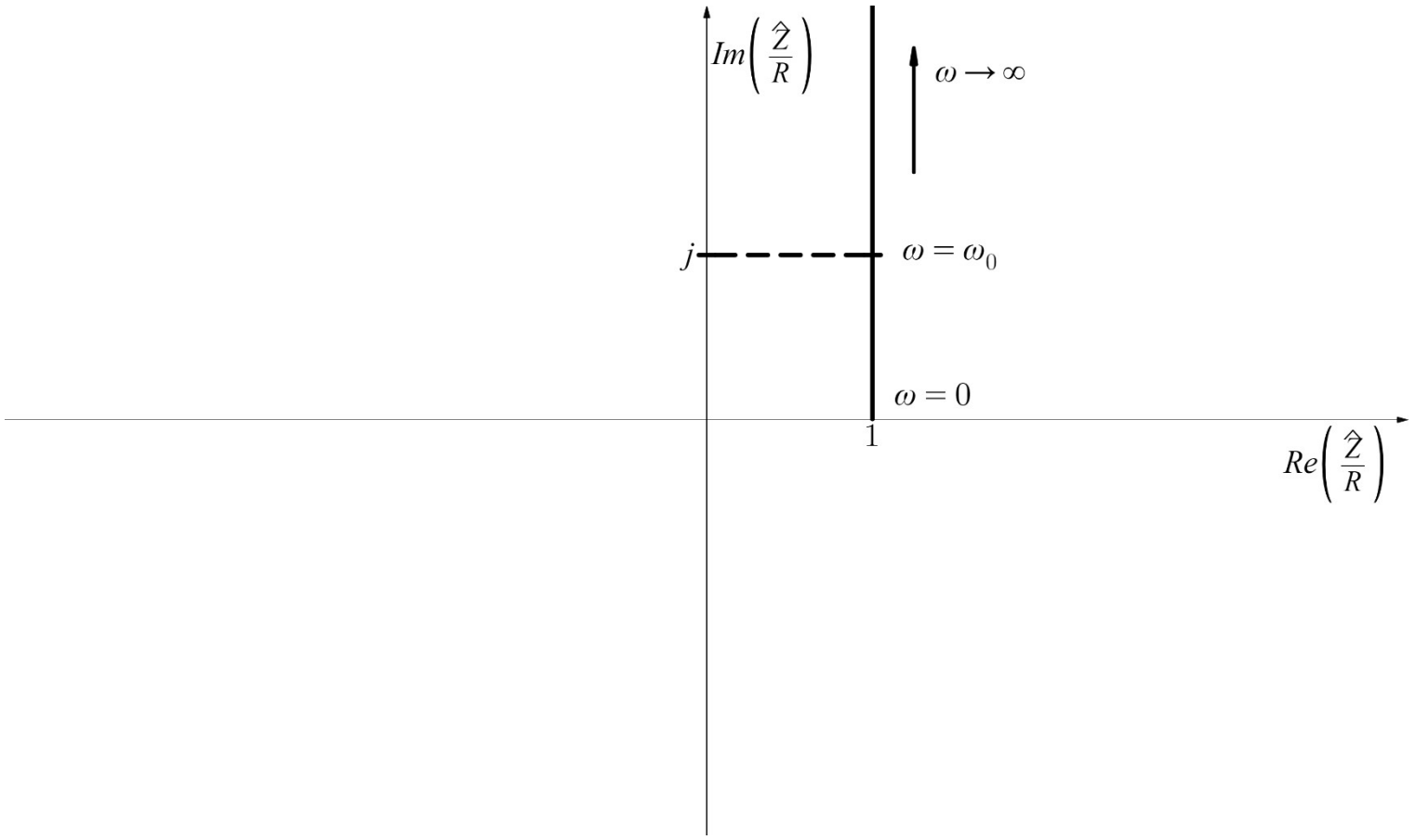
$$\tau_0 = \frac{L}{R} \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{\frac{L}{R}} = \frac{R}{L} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{R}{2\pi L}$$

$$\frac{\hat{Z}}{R} = \frac{R + j\omega L}{R} = 1 + j\omega\frac{L}{R} = 1 + j\omega\tau_0$$

$$\hat{Z}_{dB} = 20\log\left|\frac{\hat{Z}}{R}\right| = 20\log|1 + j\omega\tau_0|$$

$$\varphi = \arg\left(\frac{\hat{Z}}{R}\right) = \arg(1 + j\omega\tau_0) = \operatorname{arctg}(\omega\tau_0)$$





b) paralel

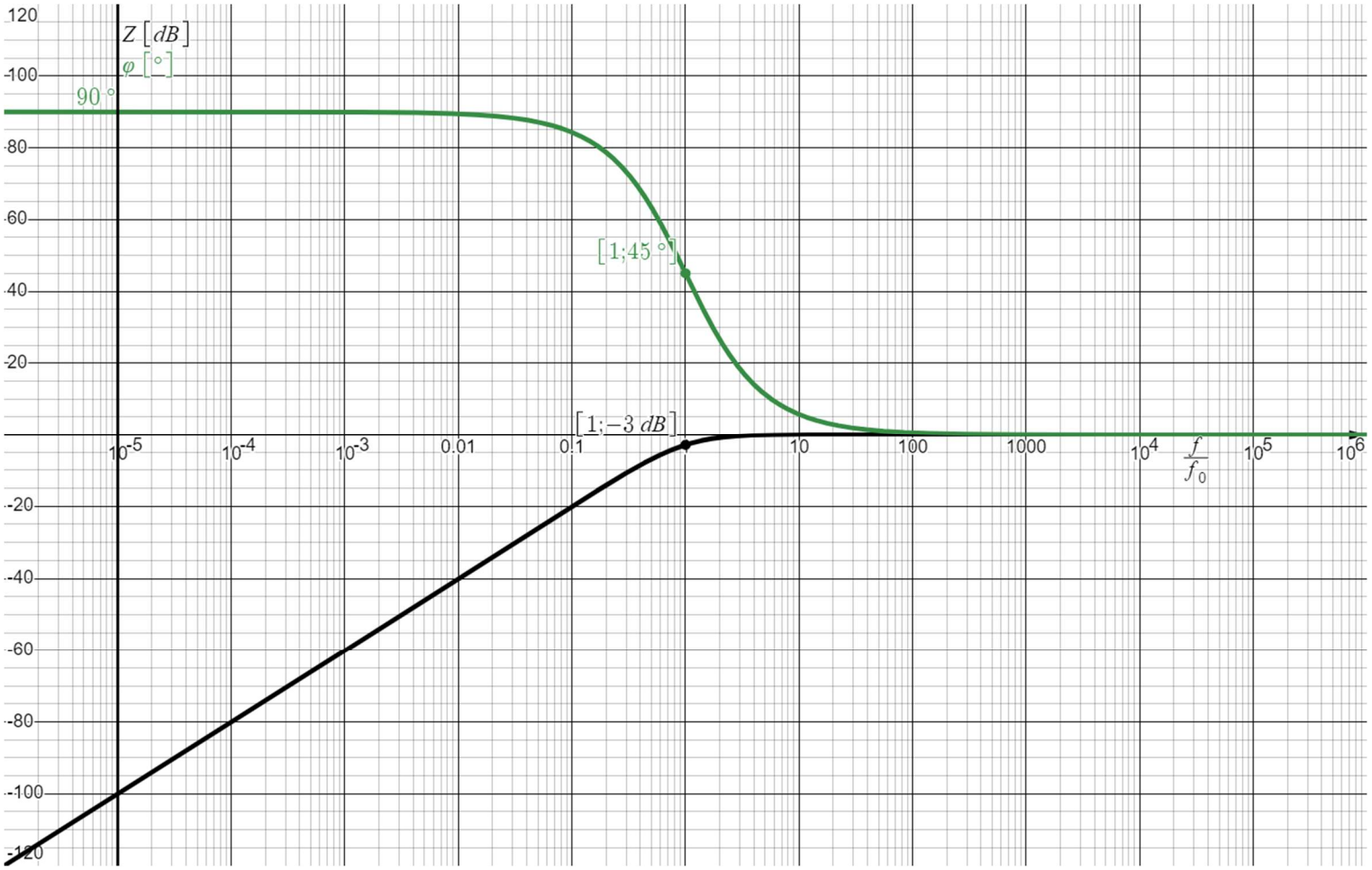
$$\hat{Z} = \frac{j\omega RL}{R + j\omega L}$$

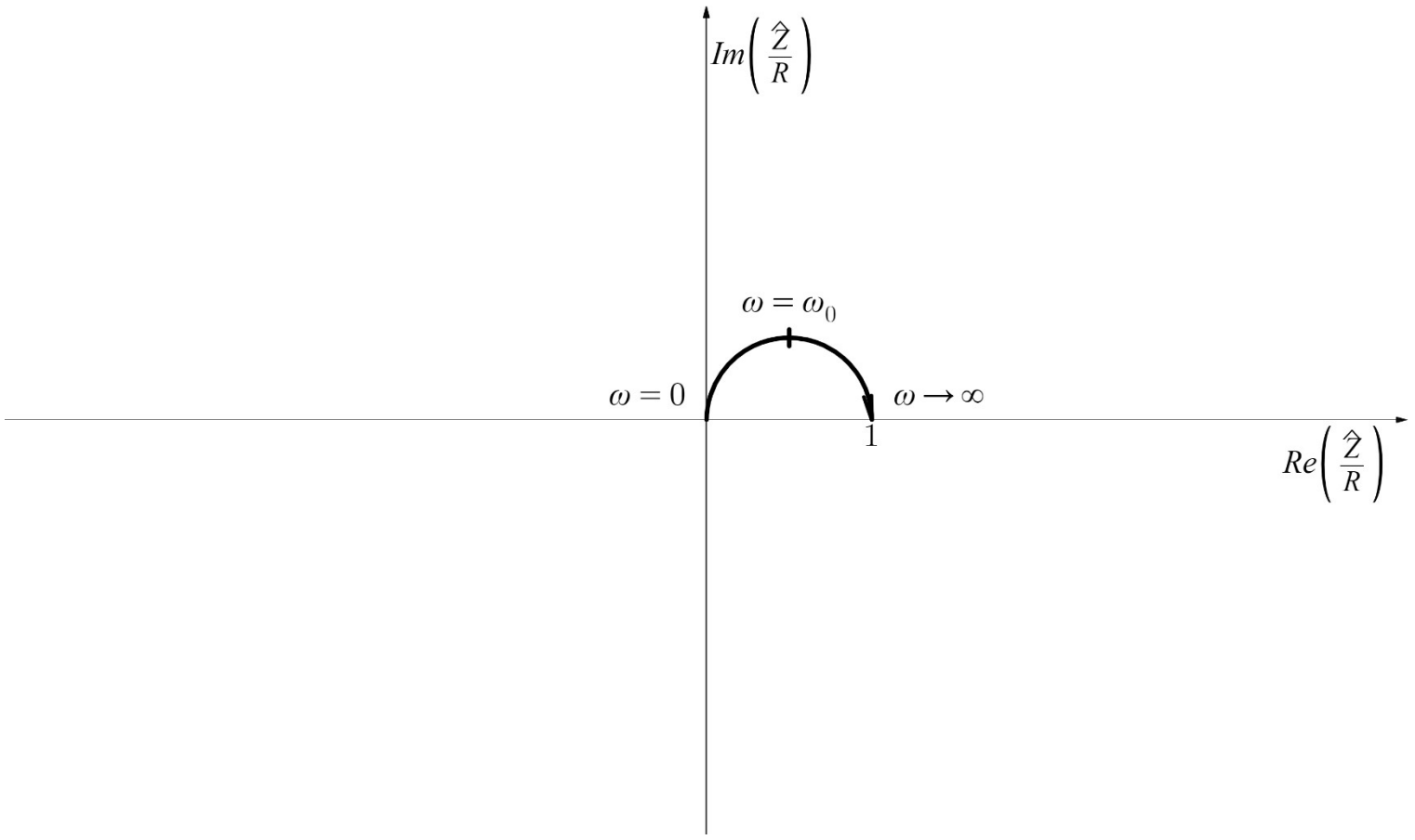
$$\tau_0 = \frac{L}{R} \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{\frac{L}{R}} = \frac{R}{L} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{R}{2\pi L}$$

$$\frac{\hat{Z}}{R} = \frac{1}{R}\frac{j\omega RL}{R + j\omega L} = \frac{1}{R}\frac{j\omega RL}{R\left(1 + j\omega\frac{L}{R}\right)} = \frac{1}{R}\frac{j\omega L}{\left(1 + j\omega\frac{L}{R}\right)} = \frac{j\omega\frac{L}{R}}{1 + j\omega\frac{L}{R}} = \frac{j\omega\tau_0}{1 + j\omega\tau_0}$$

$$\hat{Z}_{dB} = 20\log\left|\frac{\hat{Z}}{R}\right| = 20\log\left|\frac{j\omega\tau_0}{1 + j\omega\tau_0}\right|$$

$$\varphi = \arg\left(\frac{\hat{Z}}{R}\right) = \arg\left(\frac{j\omega\tau_0}{1 + j\omega\tau_0}\right) = \arg(j\omega\tau_0) - \arg(1 + j\omega\tau_0)$$
$$= \frac{\pi}{2} - \operatorname{arctg}(\omega\tau_0)$$


Z [dB]
φ [°]
90 °
[1;45 °]
[1;−3 dB]
120
100
80
60
40
20
-20
-40
-60
-80
-100
-120
10-5
10-4
10-3
0.01
0.1
10
100
1000
10^4
10^5
10^6
f/f0


Im(Ẑ/R)
ω = ω0
ω = 0
ω → ∞
1
Re(Ẑ/R)

C) RLC

a) série

$$\hat{Z} = R + j\omega L + \frac{1}{j\omega C} = R + j\omega L - \frac{j}{\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\mathrm{Im}(\hat{Z}) = 0$$

$$\omega L - \frac{1}{\omega C} = 0$$

$$\omega_0 L = \frac{1}{\omega_0 C} \implies \omega_0^2 = \frac{1}{LC} \implies \omega_0 = \frac{1}{\sqrt{LC}} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

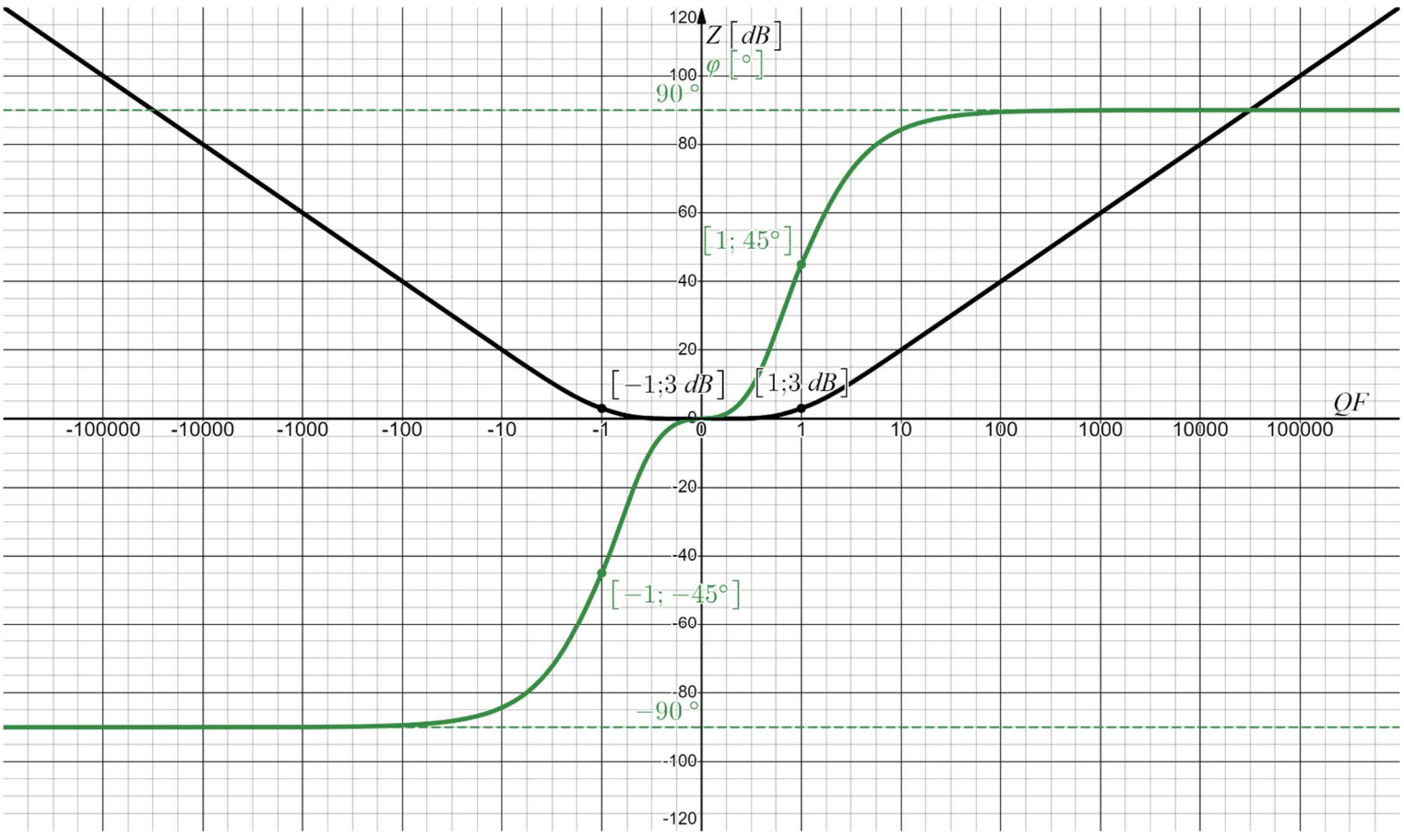
$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}$ ... činitel jakosti, $F = \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}$ ... poměrné rozladění

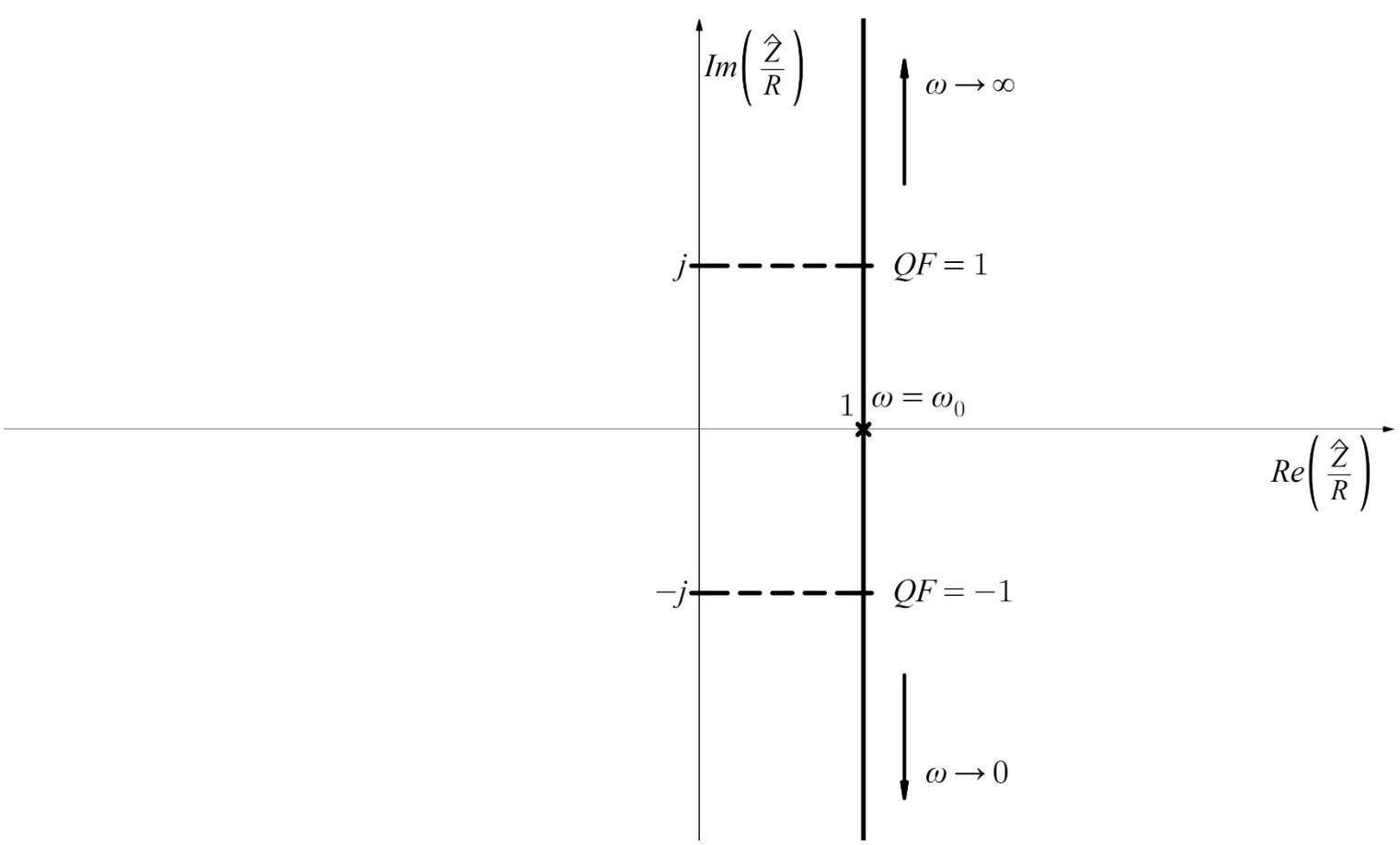
$$\frac{\hat{Z}}{R} = \frac{R + j\left(\omega L - \frac{1}{\omega C}\right)}{R} = 1 + j\left(\omega\frac{L}{R} - \frac{1}{\omega RC}\right) = 1 + j\left(\omega\frac{L}{R}\frac{\omega_0}{\omega_0} - \frac{1}{\omega RC}\frac{\omega_0}{\omega_0}\right)$$

$$= 1 + j\left(\omega\frac{L}{R}\frac{\omega_0}{\omega_0} - \frac{1}{\omega RC}\frac{\omega_0}{\omega_0}\right) = 1 + j\left(\frac{\omega}{\omega_0}Q - \frac{\omega_0}{\omega}Q\right) = 1 + jQF$$

$$\hat{Z}_{dB} = 20\log\left|\frac{\hat{Z}}{R}\right| = 20\log|1 + jQF|$$

$$\varphi = \arg\left(\frac{\hat{Z}}{R}\right) = \arg(1 + jQF) = \mathrm{arctg}(QF)$$





b) paralel

$$\hat{Z} = \frac{1}{\frac{1}{R} + \frac{1}{j\omega L} + j\omega C} = \frac{1}{\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)}$$

$$\mathrm{Im}\left(\hat{Z}\right) = 0$$

$$\omega C - \frac{1}{\omega L} = 0$$

$$\omega_0 C = \frac{1}{\omega_0 L} \implies \omega_0^2 = \frac{1}{LC} \implies \omega_0 = \frac{1}{\sqrt{LC}} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

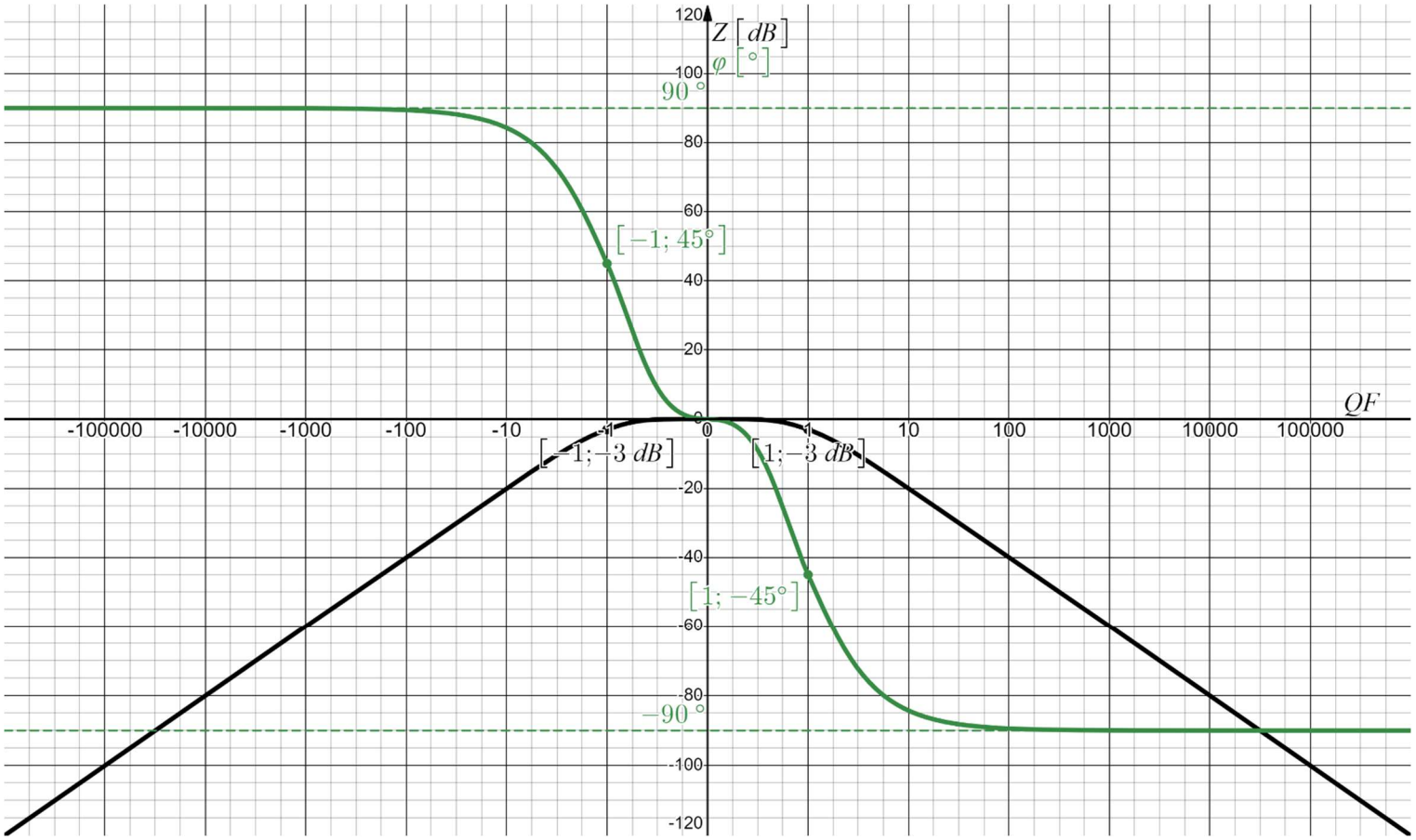
$Q = \frac{R}{\omega_0 L} = \omega_0 RC \ldots$ činitel jakosti, $F = \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \ldots$ poměrné rozladění

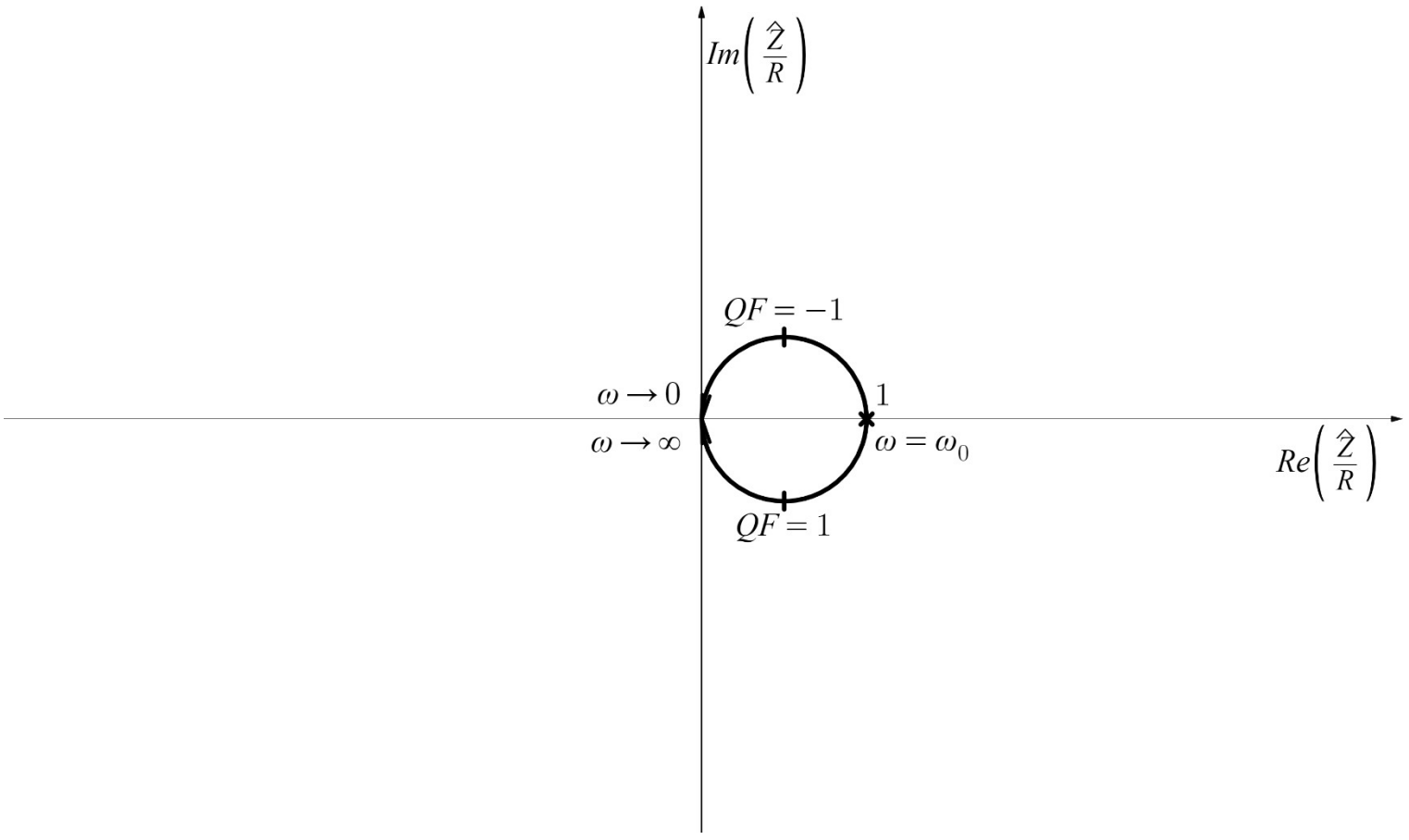
$$\frac{\hat{Z}}{R} = \frac{1}{R\left(\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)\right)} = \frac{1}{1 + j\left(\omega RC - \frac{R}{\omega L}\right)}$$

$$= \frac{1}{1 + j\left(\omega RC\frac{\omega_0}{\omega_0} - \frac{R}{\omega L}\frac{\omega_0}{\omega_0}\right)} = \frac{1}{1 + j\left(\frac{\omega}{\omega_0}Q - \frac{\omega_0}{\omega}Q\right)} = \frac{1}{1 + jQF}$$

$$\hat{Z}_{dB} = 20\log\left|\frac{\hat{Z}}{R}\right| = 20\log\left|\frac{1}{1 + jQF}\right|$$

$$\varphi = \arg\left(\frac{\hat{Z}}{R}\right) = \arg\left(\frac{1}{1 + jQF}\right) = \arg(1) - \arg(1 + jQF) = 0 - \mathrm{arctg}(QF)$$


Z [dB]
φ [°]
90 °
[−1; 45°]
QF
[−1; −3 dB]
[1; −3 dB]
[1; −45°]
−90 °
120
100
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-40
-60
-80
-100
-120
-100000
-10000
-1000
-100
-10
-1
0
1
10
100
1000
10000
100000


Im(Ẑ/R)
QF = −1
ω → 0
1
ω → ∞
ω = ω0
Re(Ẑ/R)
QF = 1